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ANTI-LIGATURE HINGE

Abstract

An anti-ligature door system that utilizes an anti-ligature hinge to operatively couple the door to the door frame. The anti-ligature hinge has an assembled leaf portion formed from a pair of leaves that are operatively coupled together using one or more pins. The anti-ligature cap that is operatively coupled to the leaf portion has a body with one or more projections extending therefrom. The one or more projections may comprise a leaf projection that is configured to be located between the first leaf and the second leaf and operatively coupled to the first leaf or the second leaf. The one or more projections may further comprise a knuckle projection that is configured to be operatively coupled to a knuckle of the first leaf or the second leaf, such as by being inserted into a knuckle aperture formed by the inner surface of the knuckle.

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Background/Summary

CROSS REFERENCE AND PRIORITY CLAIM UNDER 35 U.S.C. § 119 [0001] The present application for a patent claims priority to U.S. patent application Ser. No. 18/125,964 entitled “Anti-Ligature Hinge,” filed on Mar. 24, 2023, now U.S. Pat. No. 12,264,520, which claims priority to U.S. Provisional Patent Application Ser. No. 63/328,865 entitled “Anti-Ligature Hinge,” filed on Apr. 8, 2022, all of which are assigned to the assignees hereof and hereby expressly incorporated by reference herein.

FIELD
[0002] The present disclosure generally relates to anti-ligature hinges, and in particular a hinge cap for the anti-ligature hinge that has improved anti-ligature features.
BACKGROUND
[0003] Facilities, including medical facilities may require the use of anti-ligature products in order to protect users (e.g., patients, or the like) at the facility (e.g., visiting, outpatient, in-patient, residents, or the like). Traditional door hinges may include ligature features on which a user could be harmed. Products, including door hinges, may be improved through the use of anti-ligature products.
SUMMARY
[0004] Embodiments of the disclosure relate to an anti-ligature door system that utilizes an anti-ligature hinge to operatively couple the door to the door frame. The anti-ligature hinge may comprise an assembled leaf portion comprising a pair of leaves that are operatively coupled together using one or more hinge pins. The leaf portion may comprise an upper end (e.g., proximal end, or the like) and a lower end (e.g., a distal end, or the like). The anti-ligature hinge may be a continuous hinge, and as such may span the entire height, or a majority thereof, of the door and/or door frame. The anti-ligature hinge may comprise at least one anti-ligature cap that is operatively coupled to the upper end and/or the lower end of the assembled leaf portion. As such, in some embodiments an anti-ligature cap may be operatively coupled to both the upper end and lower end of the assembled leaf portion of the anti-ligature hinge.
[0005] The anti-ligature cap may comprise a body with one or more projections. The one or more projections may comprise a leaf projection (otherwise described as a web projection) that is configured to be located between the first leaf and the second leaf and operatively coupled to the first leaf or the second leaf. The one or more projections may further comprise a knuckle projection (otherwise described as pin projection), which is configured to be operatively coupled to a knuckle of the first leaf or the second leaf, such as by being inserted into a knuckle aperture formed by the inner surface of the knuckle. The one or more projections may be operatively coupled to the first leaf and/or the second leaf through the use of a connector (e.g., weld, fastener, crimping, button punch, interference fit, or the like). The body of the anti-ligature cap may be shaped such that it does not include any sharp edges and may be angled downwardly such that objects cannot be hung on the surfaces of the anti-ligature cap. Moreover, when installed the anti-ligature cap may be flush with the leaf portion of the hinge.
[0006] One embodiment of the present disclosure is an anti-ligature hinge. The anti-ligature hinge comprises a first leaf, a second leaf operatively coupled to the first leaf, and an anti-ligature cap. The anti-ligature cap comprises a body and one or more cap projections operatively coupled to the body or one or more cap recesses formed by the body. The one or more cap projections or the one

or more cap recesses are operatively coupled to the first leaf or the second leaf.

[0007] In further accord with embodiments, the anti-ligature cap comprises the one or more cap projections, and wherein the one or more cap projections comprise a leaf projection, wherein the leaf projection extends between the first leaf and the second leaf and is operatively coupled to the first leaf or the second leaf through a leaf projection connector.

[0008] In other embodiments, the leaf projection connector comprises a weld, a fastener, or a crimped connector.

[0009] In still other embodiments, the first leaf and the second leaf comprise a web, and a plurality of knuckles having apertures therethrough. A pin operatively couples the first leaf and the second leaf through the plurality of knuckles of the first leaf and the second leaf.

[0010] In yet other embodiments, the anti-ligature cap comprises the one or more cap projections, and wherein the one or more cap projections comprise a knuckle projection, wherein the knuckle projection extends into at least one knuckle of the plurality of knuckles of the first leaf or the second leaf.

[0011] In other embodiments, the anti-ligature cap comprises the one or more cap recesses formed by the body, wherein the one or more cap recesses are configured to receive one or more hinge projections operatively coupled to the first leaf or the second leaf.

[0012] In further accord with embodiments, the first leaf or the second leaf have an upper end with a notched section.

[0013] In other embodiments, the notched section comprises a beveled surface that extends from a knuckle of the first leaf or the second leaf and into at least a portion of the upper end of the first leaf or the second leaf.

[0014] In still other embodiments, the beveled surface has an angle from the horizontal plane of an upper door edge that ranges from 10 to 80 degrees, inclusive.

[0015] In yet other embodiments, the body of the anti-ligature cap comprises a bottom surface. The one or more cap projections extend from the bottom surface of the body, or the one or more cap recesses are formed within the bottom surface of the body. At least a portion of the bottom surface engages with the notched section.

[0016] In further accord with embodiments, the body of the anti-ligature cap comprises a top surface comprising of an exposed portion and a frame portion. The frame portion is located within a door frame when the anti-ligature hinge is installed.

[0017] In other embodiments, the exposed portion diverges from the frame portion outwardly and downwardly from a horizontal plane of the frame portion.

[0018] In yet other embodiments, the exposed portion diverges from the frame portion at an anti-ligature angle from the horizontal plane of the frame portion that ranges from 10 to 80 degrees, inclusive.

[0019] In still other embodiments, the body of the anti-ligature cap comprises a first side cap surface and a second side cap surface. The first side cap surface is flush with a first outer surface of the first leaf and the second side cap surface is flush with a second outer surface of the second leaf.

[0020] In other embodiments, the anti-ligature cap is a cast anti-ligature cap.

[0021] In still other embodiments, the anti-ligature hinge is a continuous anti-ligature hinge.

[0022] Another embodiment of the present disclosure is a door system. The door system comprises a door, a door frame, and an anti-ligature hinge. The anti-ligature hinge comprises a first leaf, a second leaf operatively coupled to the first leaf, and an anti-ligature cap. The anti-ligature cap comprises a body and one or more cap projections operatively coupled to the body, or one or more cap recesses formed by the body. The one or more cap projections or the one or more cap recesses are operatively coupled to the first leaf or the second leaf.

[0023] In further accord with embodiments, the anti-ligature cap comprises the one or more cap projections, and wherein the one or more cap projections comprise a leaf projection, wherein the leaf projection extends between the first leaf and the second leaf and is operatively coupled to the

first leaf or the second leaf through a leaf projection connector.
[0024] In other embodiments, the anti-ligature cap comprises the one or more cap projections, and wherein the one or more cap projections comprise a knuckle projection, wherein the knuckle projection extends into at least one knuckle of a plurality of knuckles of the first leaf or the second leaf.
[0025] Another embodiment of the present disclosure comprises a method of installing an anti-ligature hinge. The method comprises assembling one of a first leaf or one of a second leaf to a door frame, assembling another of the first leaf or the second leaf to a door, and assembling an anti-ligature cap to the first leaf or the second leaf.
[0026] To the accomplishment the foregoing and the related ends, the one or more embodiments comprise the features hereinafter described and particularly pointed out in the claims. The following description and the annexed drawings set forth certain illustrative features of the one or more embodiments. These features are indicative, however, of but a few of the various ways in which the principles of various embodiments may be employed, and this description is intended to include all such embodiments and their equivalents.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0027] Having thus described embodiments of the invention in general terms, reference will now be made to the accompanying drawings.
[0028] FIG. **1**A is a front view of a door system with a continuous anti-ligature hinge, in accordance with embodiments of the present disclosure.
[0029] FIG. **1**B is a first side view of a continuous anti-ligature hinge with an anti-ligature cap on the upper end and an anti-ligature cap on the lower end, in accordance with embodiments of the present disclosure.
[0030] FIG. **1**C is a second side view of a continuous anti-ligature hinge with an anti-ligature cap on the upper end and an anti-ligature cap on the lower end, in accordance with embodiments of the present disclosure.
[0031] FIG. **2** is an enlarged perspective view of a door system with a continuous anti-ligature hinge, in accordance with embodiments of the present disclosure.
[0032] FIG. **3** is an enlarged perspective view of the door system with the continuous anti-ligature hinge of FIG. **2** with the door removed, in accordance with embodiments of the present disclosure.
[0033] FIG. **4** is an enlarged perspective view of the door system with the continuous anti-ligature hinge of FIG. **2** with the door and anti-ligature cap removed, in accordance with embodiments of the present disclosure.
[0034] FIG. **5** is an enlarged perspective view of the door system with the continuous anti-ligature hinge of FIG. **2** with the door and second hinge leaf removed, in accordance with embodiments of the present disclosure.
[0035] FIG. **6** is an enlarged perspective view of a continuous anti-ligature hinge, in accordance with embodiments of the present disclosure.
[0036] FIG. **7** is an enlarged perspective view of the continuous anti-ligature hinge of FIG. **6** with the second hinge leaf removed, in accordance with embodiments of the present disclosure.
[0037] FIG. **8**A is an enlarged perspective view of the continuous anti-ligature hinge of FIG. **6** with the anti-ligature cap removed, in accordance with embodiments of the present disclosure.
[0038] FIG. **8**B is an enlarged perspective view of the continuous anti-ligature hinge of FIG. **6** with the second hinge leaf and anti-ligature cap removed, in accordance with embodiments of the present disclosure.
[0039] FIG. **9** is a perspective view of an anti-ligature cap for an anti-ligature hinge, in accordance

with embodiments of the present disclosure.
[0040] FIG. **10**A is a first side view of an anti-ligature cap for an anti-ligature hinge, in accordance with embodiments of the present disclosure.
[0041] FIG. **10**B is a second side view of an anti-ligature cap for an anti-ligature hinge, in accordance with embodiments of the present disclosure.
[0042] FIG. **11**A is a top view of an anti-ligature cap for an anti-ligature hinge, in accordance with embodiments of the present disclosure.
[0043] FIG. **11**B is a bottom view of an anti-ligature cap for an anti-ligature hinge, in accordance with embodiments of the present disclosure.
[0044] FIG. **12**A is a front view of an anti-ligature cap for an anti-ligature hinge, in accordance with embodiments of the present disclosure.
[0045] FIG. **12**B is a rear view of an anti-ligature cap for an anti-ligature hinge, in accordance with embodiments of the present disclosure.
[0046] FIG. **13** is a perspective view of an anti-ligature cap for an anti-ligature hinge, in accordance with embodiments of the present disclosure.
[0047] FIG. **14**A is a first side view of an anti-ligature cap for an anti-ligature hinge, in accordance with embodiments of the present disclosure.
[0048] FIG. **14**B is a second side view of an anti-ligature cap for an anti-ligature hinge, in accordance with embodiments of the present disclosure.
[0049] FIG. **15**A is a top view of an anti-ligature cap for an anti-ligature hinge, in accordance with embodiments of the present disclosure.
[0050] FIG. **15**B is a bottom view of an anti-ligature cap for an anti-ligature hinge, in accordance with embodiments of the present disclosure.
[0051] FIG. **16**A is a front view of an anti-ligature cap for an anti-ligature hinge, in accordance with embodiments of the present disclosure.
[0052] FIG. **16**B is a rear view of an anti-ligature cap for an anti-ligature hinge, in accordance with embodiments of the present disclosure.
[0053] FIG. **17** illustrates a perspective top view of an anti-ligature cap for an anti-ligature hinge, in accordance with embodiments of the present disclosure.
[0054] FIG. **18** illustrates a perspective bottom view of an anti-ligature cap for an anti-ligature hinge, in accordance with embodiments of the present disclosure.
[0055] FIG. **19**A is a perspective view of an anti-ligature cap being assembled to an assembled leaf portion, in accordance with embodiments of the present disclosure.
[0056] FIG. **19**B is a perspective view of an anti-ligature hinge with the hinge leaves in a closed position, in accordance with embodiments of the present disclosure.
[0057] FIG. **19**C is a perspective view of an anti-ligature hinge with the hinge leaves in an open position, in accordance with embodiments of the present disclosure.
[0058] FIG. **20**A is a perspective view of an anti-ligature hinge having an anti-ligature cap with a cap leaf, in accordance with embodiments of the present disclosure.
[0059] FIG. **20**B is a perspective view of the anti-ligature hinge having the anti-ligature cap with the cap leaf of FIG. **20**A with the second leaf removed, in accordance with embodiments of the present disclosure.
[0060] FIG. **20**C is a perspective view of the anti-ligature cap with the cap leaf of FIG. **20**A, in accordance with embodiments of the present disclosure.
[0061] FIG. **21** is a method of manufacturing and installing the anti-ligature hinge, in accordance with embodiments of the present disclosure.

DETAILED DESCRIPTION

[0062] The following detailed description teaches specific example embodiments of the invention; however, other embodiments of the invention do not depart from the scope of the present invention. The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms "a", "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "includes" and/or "including" when used herein, specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof.

[0063] Embodiments of the invention will be described with respect to FIGS. **1**A through **20**C illustrating an anti-ligature hinge **100**, which in some embodiments is a continuous anti-ligature hinge **100** that provides improved features for anti-ligature hinges. The anti-ligature hinge **100** may be a continuous anti-ligature hinge (e.g., extending along a portion of, the majority of, or the entire length of the door edge of the door **10** and/or the door frame **20**), and is operatively coupled to the door **10** and the door frame **20** for pivotal movement of the door **10** relative to the door frame **20** between a closed position and one or more open positions (e.g., including a fully open position). As illustrated in FIGS. **1**B and **1**C, and as will be described in further detail herein, the anti-ligature hinge **100** may comprise an assembled leaf portion **102** comprising a pair of leaves **130**, **150** that are operatively coupled together using one or more hinge pins **170**. As illustrated in FIGS. **1**A through **1**C, the leaf portion **102** may comprise an upper end **104** (e.g., proximal end, or the like) and a lower end **106** (e.g., a distal end, or the like). The anti-ligature hinge **100** may comprise at least one anti-ligature cap **200** that is operatively coupled to the upper end **104** of the assembled leaf portion **102** and/or the lower end **104** of the assembled leaf portion **102** (e.g., in some embodiments, to both ends **104**, **106**). The anti-ligature cap **200** may comprise a body **210** with one or more projections **250**. The one or more projections **250** may comprise a leaf projection **260** that is configured to be located between the first leaf **130** and the second leaf **150** and operatively coupled to the first leaf **130** or the second leaf **150**. The one or more projections **250** may further comprise a knuckle projection **270** that is configured to be operatively coupled to a knuckle of the first leaf **130** or the second leaf **150**, such as by being inserted into a knuckle aperture formed by the inner surface of the knuckle. The one or more projections **250** may be operatively coupled to the first leaf **130** and/or the second leaf **150** through the use of a connector (e.g., weld, fastener, crimping, button punch, interference fit, or the like). It should be understood that the fastener may be any type of fastener, such as, a screw, bolt, rivet, pin, or other like fastener. The body of the anti-ligature cap **200** may be shaped such that it does not include any sharp edges and may be angled downwardly such that objects cannot be hung on the surfaces of the anti-ligature cap **200**. Moreover, when installed the anti-ligature cap **200** may be flush with the leaf portion **102** of the hinge.

[0064] FIG. **1**A illustrates a door system **1** that utilizes the anti-ligature hinge **100** illustrated and described with respect to FIGS. **1**B through **20**C. As illustrated in FIG. **1**, the door system **1** may comprise a door **10** operatively coupled to a door frame **20**, which is operatively coupled to a wall (e.g., an opening in a wall). The door frame **20** may comprise three (3) portions including an upper portion **24** disposed adjacent an upper end (e.g., header) of a door opening, and two side portions **26**, **28** disposed along either edge of the door opening, with one side portion **26** being on the hinge side of the door **10**, and the opposite side portion **28** being on the latch side of the door **10**. The door frame portions **24**, **26**, **28** may be secured to each other and/or an adjoining structure by frame connectors (e.g., clips, tabs, fasteners, or the like). The door **10** may be hung otherwise conventionally within the door opening by the anti-ligature hinge **100**, such as through hinge connectors **30**. The connectors **30** may comprise fasteners that secure the anti-ligature hinge **100** through leaf apertures in one leaf to a door frame **20** (e.g., to one or more hinge reinforcements in the door frame **20**) and/or in another leaf to an edge of the door **10** (e.g., to a side edge that may or may not have hinge reinforcements in the door **10**). Other types of hinge connectors **30**, such as welds, non-removeable fasteners, or the like may be used to operatively couple the anti-ligature hinge **100** to the door **10** and/or door frame **20**. It should be understood that the door **10** and door frame **20** may be any type of conventional door and door frame of any size and shape, and made of any type of material (e.g., wood, metal hollow, metal with a core, plastic, composite, or the like, or combinations thereof).

[0065] As illustrated in FIGS. **3** through **5**, the anti-ligature hinge **100** may comprise a leaf portion **102** comprising a pair of leaves **130**, **150** that are operatively coupled together using one or more pins **170**. The leaf portion **102** may comprise an upper end **104** (e.g., proximal end, or the like) and a lower end **106** (e.g., a distal end, or the like). Moreover, the anti-ligature hinge **100** may comprise at least one anti-ligature cap **200** that is operatively coupled to one or more of the ends **104**, **106** the leaf portion **102**. As illustrated in FIG. **2**, the anti-ligature cap **200** is operatively coupled to the upper end **104** of the leaf portion **102**. It should be understood that the assembly of the anti-ligature cap **200** is described herein with respect to the upper end **102** of the leaf portion **102** of the anti-ligature hinge **100**; however, it should be understood that the anti-ligature cap **200** may be assembled to the lower end **102** of the leaf portion of the anti-ligature hinge **100** in the same or similar way.

[0066] As illustrated throughout the figures, the leaf portion **102** may comprise a first leaf **130** and a second leaf **150**. Each of the two leaves **102** may have webs that are operatively coupled to knuckles. For example, the first leaf **130** may comprise a first web **134** operatively coupled to a plurality of first knuckles **136**, while the second leaf **150** may comprise a second web **154** operatively coupled to a plurality of second knuckles **156**. The plurality of knuckles **134**, **154** may be spaced apart such that the plurality of first knuckles **136** may interlock with the plurality of second knuckles **156** (e.g., in an alternating pattern, non-uniform pattern, or the like). The plurality of knuckles may have first apertures **138** and second apertures **158** that are formed by the inner surfaces of the plurality of knuckles **136**, **156**, and are aligned when the plurality of knuckles **136**, **156** are interlocking. The plurality of first knuckles **136** are operatively coupled to the plurality of second knuckles **156** through the use of one or more hinge pins **170** that extend through at least a portion of the first apertures **138** and the second apertures **158**.

[0067] Like the leaf portion **102**, the first leaf **130** may have a first upper end **140** and a first lower end **142**, while the second leaf **150** may have a second upper end **160** and a second lower end **162**. As illustrated in FIGS. **7**-**8**B, the first upper end **140** and/or the second upper end **160** may have at least a portion that is a notched section **180**. The notched section **180** may be any type of shape, but in some embodiments the notched section may be rectangular, square, triangular, stepped, curved (e.g., semi-circular, semi-oval), parabolic, hyperbolic, or the like, and moreover, may be uniform or non-uniform in shape. In some embodiments the notched section **180** may be a bevel **182** of uniform or non-uniform shape. As illustrated in the figures, the bevel **182** may be defined as a top bevel that is uniform (e.g., planer). In some embodiments, the notched section **180** only occurs in the webs **134**, **154**, but in other embodiments, as illustrated in FIGS. **8**A and **8**B, the notched section **180** may be located in the webs **134**, **154** and a knuckle (e.g., a first knuckle **136** or a second knuckle **156** depending on the hinge configuration) of the anti-ligature hinge **100**.

[0068] As illustrated in FIGS. **9** through **16**B, the anti-ligature cap **200** may comprise a body **210** with one or more projections **250**. The body of the anti-ligature cap **200** may comprise a top surface **212** having a frame section **214** and an exposed section **216**. It should be understood that the frame section **214** may be generally planer and may be hidden when assembled to the leaf portion **102** of the anti-ligature hinge **100** when the anti-ligature hinge **100** is assembled to the door **10** and door frame **20**. The exposed section **216** of the top surface **212** of the anti-ligature cap **200** may diverge away from and extend downwardly towards the floor when the anti-ligature cap **200** is installed on the upper end **104** of the leaf portion **102** of the anti-ligature hinge **100**. However, the exposed section **216** may diverge away from and extend upwardly towards the ceiling when the anti-ligature cap **200** is installed on the lower end **106** of the leaf portion **102** of the anti-ligature hinge **100**. In some embodiments, the diverging exposed section **216** may have an angle of 10, 20, 30, 40, 45, 50, 60, 70, 80, or a range of angles of any of the foregoing that fall within, outside, or overlap the angles above, with respect to a horizontal plane as noted by reference A in FIG. **10**A. In some embodiments, the angle A may range between 10 and 80 degrees, and in particular embodiments between 30 and 60 degrees or between 40 and 50 degrees. It should be understood that an angle of a beveled notched section **180** of the leaf portion **102** may be the same as or different than the angle A described above.

[0069] As illustrated in FIGS. **9** through **16**B, the exposed section **216** of the anti-ligature cap **200** may further comprise a generally cylindrical shape **218** that is flush with at least a portion of the outer surface of the knuckle **136** to which the anti-ligature cap **200** is operatively coupled during installation. Moreover, the anti-ligature cap **200** may further comprise a first side cap surface **220** and a second side cap surface **222**. When installed, as illustrated in FIG. **6**, the first side cap surface **220** may be flush with an outer surface **132** of the first leaf **130**, and the second side cap surface **222** may be flush with an outer surface **152** of the second leaf **150**. In this way, the frame section **214** of the anti-ligature hinge **100** may be located between the edge surface of the door **10** and the jamb surface of the door frame **20** when the anti-ligature hinge **100** is installed and the door **10** of the door system **1** is closed, as illustrated in FIG. **2**.

[0070] As illustrated in FIGS. **11**B-**12**B and **15**B-**16**B, the anti-ligature cap **200** may further comprise a bottom surface **240** from which the one or more projections **250** extend. As such, the bottom surface may comprise a knuckle shoulder **242**, a first leaf shoulder **244**, and a second leaf shoulder **246**. It should be understood that the knuckle shoulder **242** may engage with a knuckle **134**, **154** of the first leaf **130** or the second leaf **150**. Furthermore, it should be understood that the first leaf shoulder **244** may engage with the notched section **180** of the end of the first leaf **130** and the second leaf shoulder **246** may engage with the notched section **180** of the end of the second leaf **150**. Depending on the shape of the notched section **180** of the anti-ligature hinge **100**, the anti-ligature cap **200** may have multiple shoulders that are configured to engage with different surfaces of the notched section **180** (e.g., square, curved, stepped, or the like surface of the notched section **180** of the end of the leaf portion **102** of the anti-ligature hinge **100**. While the anti-ligature cap **200** is illustrated as tapering, and only extending over a portion of the upper end **104** of the leaf portion **102**, in some embodiments the length of the assembled leaf portion **102** (or a first leaf **130** or a second leaf **150** thereof) may be shortened, have a notched section **180**, or the like such that at least a portion of the anti-ligature cap **200** extends over the entire upper end **104** of the leaf portion **102**.

[0071] As discussed herein, the one or more projections **250** of the anti-ligature cap **200** may comprise a leaf projection **260** and a knuckle projection **270**. The leaf projection **260** may extend from the bottom surface **240** of the anti-ligature cap **200**. The leaf projection **260** may be configured to be located between the inner surface of the first web **134** of the first leaf **130** and the inner surface of the second web **154** of the second leaf **150**. In some embodiments, the leaf projection **260** may have a thickness that is less than or equal to the size of the space between the first web **134** and the second web **154** when the door **10** of the door system **1** is in a closed position. In some embodiments, the leaf projection **260** may have an angled thickness that is converging or diverging (e.g., a draft angle, a tapered thickness, or the like), a non-uniform thickness, a stepped configuration with different thickness, or other like configuration. The leaf projection **260** is illustrated as having a triangular shape; however, it should be understood that the leaf projection **260** may have any type of shape (e.g., semi-circular, square, rectangular, or the like) that allows the leaf projection **260** to be located between the first leaf **130** and the second leaf **150**. Furthermore, while FIGS. **9-12**B illustrate that the leaf projection **260** has a uniform shape (e.g., triangular), it should be understood that the leaf projection **260** may have a notched portion **262** that provides a non-uniform shape. For example, as illustrated in FIGS. **13-16**B, the leaf projection **260** may have notched section **262** that allows for the use of the anti-ligature cap **200** with multiple anti-ligature hinges **100** that have different aperture patterns for operative coupling with a door **10** and/or a door frame **20**. That is, the notch **262** may allow a connector **30** to pass through the web of the first leaf **130** and/or the second leaf **150** without interfering with the leaf projection **260**.

Moreover, as will be discussed in further detail herein, the leaf projection **260** may be operatively coupled to the inner surface of the first web **134** or the inner surface of the second web **154** through the use of one or more connectors (e.g., fasteners, welds, crimping, button punches, interference fits, or the like) depending on the installation of the anti-ligature hinge **100**.

[0072] The knuckle projection **270** (or pin projection **270**) may be configured to be operatively coupled to a knuckle **136**, **156** of the first leaf **130** or the second leaf **150**, as will be discussed in further detail below. The knuckle projection **270** is illustrated as being a pin having a generally cylindrical shape with a converging end, which is configured to be located within an aperture **138**, **158** created by the inner surface of a knuckle **134**, **154** of the first leaf **130** or the second leaf **150**. However, it should be understood that the knuckle projection **270** may have different types of shapes (e.g., square, half-circular, triangular, or the like).

[0073] While the embodiments illustrated in the figures indicate that the one or more projections **250** are extending from the anti-ligature cap **200**, it should be understood that alternatively, or additionally, the anti-ligature cap **200** may have recesses. The recesses in the anti-ligature cap **200** may receive members that are operatively coupled to the first leaf **130** and/or second leaf **150** (e.g., independent members with projections that are welded thereto) and/or projections that extend from the first leaf **130** and/or the second leaf **150** (e.g., projecting from the knuckles **136**, **156** and/or the webs **134**, **154**). As such, the anti-ligature cap **200** may be operatively coupled to the leaf portion **102** at least partially by receiving one or more projections in one or more recesses. For example, in some embodiments, as illustrated in FIGS. **17** and **18**, the knuckle projection **270**, **276** extends from a knuckle **136**, **156** of the first leaf **130** and/or the second leaf **150** (e.g., is permanently or removable operatively coupled to one or more of the knuckles **136**, **156**), and is operatively coupled to the anti-ligature cap **200** through a recess **278** located in the anti-ligature cap **200**.

[0074] FIGS. **19**A through **19**C illustrates how an anti-ligature cap **200** is assembled to the leaf portion **102** of the anti-ligature cap **200**. As illustrated in FIG. **19**A the leaf projection **260** of the anti-ligature cap **200** may be slid between the first web **134** and the second **154** and/or the knuckle projection **270** may be slid into a knuckle aperture **138**, **158** formed by a knuckle **136**, **156**. As such, as illustrated in FIG. **19**B, the anti-ligature cap **200** may be press-fit or may be removably operatively coupled to the leaf portion **102** (e.g., loosely assembled). As illustrated in FIG. **19**C, the anti-ligature cap **200** may be operatively coupled to an inner surface of a leaf web (e.g., a first leaf web **134** as illustrated, or a second leaf web **154** in a different configuration). Additionally, or alternatively, a knuckle (e.g., a first knuckle **136** as illustrated, or a second knuckle **156** in a different configuration) may be operatively coupled to the knuckle projection **270**. The connectors **280** may be spot welds **282** (e.g., as illustrated in FIG. **19**C) or may be other types of connectors **280**, such as other types of welds (e.g., butt weld, edge weld, or the like), fasteners (e.g., through the leaf projection **260**, first web **134**, and into a door frame **20** or door **20**, or the like).

[0075] FIGS. **20**A through **20**C illustrate that in other embodiments of the invention, the anti-ligature cap **200** may include a cap leaf **290** that may act as at least a portion of the first leaf **130** or the second leaf **150**. As such, in the embodiments illustrated in FIGS. **20**A through **20**C, only the second leaf **150** has the notched section **180** and the first leaf **130** is shorter than the second leaf **150**.

[0076] Regardless of the configurations illustrated in FIGS. **1**A through **20**C, the anti-ligature hinges **100** may further comprise inserts, bearings, or other components between the individual knuckles **136**, **156**, between a knuckle **136**, **156** and the anti-ligature cap **200**, or the like in order to allow the hinge to rotate more freely, to restrict play in the assembled components, or the like.

[0077] It should be understood that the anti-ligature hinge **100**, and in particular the anti-ligature cap **200** thereof, provide improved anti-ligature features, which reduce the chances that users could harm themselves. For example, the anti-ligature cap **200** has angled and/or rounded surfaces that reduce the ability for a user be to cut or hang something (e.g., shoelaces, sheets, rope, or the like) from the anti-ligature cap **200**. Moreover, the configuration of the anti-ligature cap **200** improves the case of manufacturing and assembly by eliminating the need to machine rounded and angled features into hinge. Moreover, the anti-ligature cap **200** allows for the anti-ligature hinge **100** be installed in right or left handed door that swings in or out.

[0078] The anti-ligature hinge **100** resists the ability of a flexible device to be looped around or tied off on the anti-ligature hinge **100** and/or to be jammed or wedged into any openings (e.g., since the cap **200** is flush with the hinge portion). Moreover, the anti-ligature hinge **100**, in particular the anti-ligature cap **200**, has round surfaces that resists cutting. Moreover, due to the assembly of the anti-ligature cap **200** with the leaf portion **102** of the anti-ligature hinge **100** (e.g., cap **200** located adjacent the door frame **20**, operatively coupled with one or more connectors, and/or the like), the anti-ligature cap **200** cannot be removed once installed (e.g., without dissembling the anti-ligature hinge **100** from the door frame **20** and/or removing or damaging the connectors, or the like).

[0079] The anti-ligature hinge **100**, in particular, the anti-ligature cap **200** (e.g., the top surface thereof) will at a minimum not hold a force greater than 11 lbs., and in some cases not hold a force of 10, 9, 8, 7, 6, 5, 4, 3, 2, 1, or the like lbs. or a range of any of the forgoing values. In particular embodiments, the anti-ligature cap **200** (e.g., the top surface thereof) will not support a force that is greater than 5.5 lbs. That is, for example should a lace or other flexible object be looped around the anti-ligature hinge it will release the flexible object when a force of greater than 5.5 lbs. is applied (or in other embodiments when a force between 1 and 11 lbs. described above is applied) in any direction (e.g., up, left, right, and in particular down, and/or at different angles).

[0080] FIG. **21** illustrates a method of manufacturing, shipping, and/or installing the anti-ligature hinges **100** described herein. As illustrated in block **302** of FIG. **21**, the hinge leaves **130**, **150** may be manufactured to create the desired lengths, web sizes of the webs **134**, **154**, sizes and number of knuckles **136**, **156**, or the like. It should be understood that leaves **130**, **150** of a continuous hinge may be formed, and thereafter, cut into the desired lengths. Moreover, the notch section **180** of the leaves **130**, **150** may be formed during manufacturing or may be cut into the leaves **130**, **150** of the first leaf **130** and the second leaf **150**. The materials used for the hinge **100** may include, for example, various materials such as aluminum, steel, stainless steel, plastic resin, composite (e.g., plastic resin over-molded onto a substrate of different material, or other like composites), or the like, or combinations thereof.

[0081] Block **304** of FIG. **21** further illustrates that the anti-ligature cap **200** may be manufactured. For example, anti-ligature cap **200** may be cast, machined, or the like, or combinations thereof. In particular embodiments, the anti-ligature cap **200** may be cast, which may reduce the costs associated with creating the anti-ligature features of the anti-ligature cap **200**.

[0082] FIG. **21** further illustrates in block **306** that the components of the anti-ligature hinge **200** may be shipped separately or at least partially pre-assembled. For example, the anti-ligature hinge **200** may be at least partially pre-assembled by operatively coupling a first leaf to a second leaf by assembling one or more hinge pins **170** to the plurality of interlocking knuckles **134**, **154** of the first leaf **130** and the second leaf **150**. The anti-ligature cap **200** may be pre-assembled (e.g., fully assembled) or may be sent at least partially unassembled to allow the anti-ligature hinge **100** to be assembled on site based on the door being a right hand swinging, left hand swinging, inward swinging, and/or outward swinging door **10**.

[0083] Block **308** of FIG. **18** further illustrates that an installer operatively couples the first leaf **130** to the second leaf **150** through the use of the hinge pin **170**, if the anti-ligature hinge **100** is not pre-assembled into the assembled leaf portion **102**. The installer may then operatively couple the anti-ligature cap **200** to the leaf portion **102** of the anti-ligature hinge **100** if it was not at least partially pre-assembled. The installer then operatively couples the first leaf **130** or the second leaf **150** to the door frame **20** and the opposite first leaf **130** or second leaf **150** to the door **10** through the use of connectors. For example, the connectors may be welds, fasteners, crimping, or the like.

[0084] FIG. **21** further illustrates in block **310** that an installer operatively couples the anti-ligature cap **200** to the first leaf **130** or the second leaf **150** through the use of one or more connectors, such as by spot welding a knuckle **136**, **156** to a knuckle projection **270** and/or spot welding a leaf projection **260** to a first web **134** of a first leaf **130** or a second web **154** of a second leaf **150** (e.g., depending on the installation). In other embodiments, other types of welds (e.g., edge welds, butt welds, or the like), fasteners, or the like may be used as the connector. For example, a fastener may operatively couple a leaf projection **260** to a first web **134** or a second web **154** (e.g., through web apertures thereof, such as holes that may be countersunk, or the like) and/or to the door **10** or door frame **20**.

[0085] It should be understood that while the present invention is described with respect to continuous barrel and pin anti-ligature hinges, the present invention may also be used in accordance with other types of hinges, such as non-continuous hinges, or the like. Moreover, while the anti-ligature cap **200** is illustrated as being operatively coupled at the upper end **104** of the assembled leaf portion **102**, it should be understood that an anti-ligature cap **200** may be operatively coupled to the leaf portion **102** at the bottom end **106**.

[0086] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. It will be further understood that terms used herein should be interpreted as having a meaning that is consistent with their meaning in the context of this specification and the relevant art and should not be interpreted in an idealized or overly formal sense unless expressly so defined herein. Certain terminology is used herein for convenience only and is not to be taken as a limitation on the invention. For example, words such as “top,” “bottom,” “side,” “distal,” “proximal,” “upper,” “lower,” “left,” “right,” “horizontal,” “vertical,” “upward,” “downward”, “first”, “second”, or other like terminology merely describe the configuration shown in the figures. The referenced components may be oriented in an orientation other than that shown in the drawings and the terminology, therefore, should be understood as encompassing such variations unless specified otherwise.

[0087] It will be understood that when an element is referred to as “operatively coupled” to another element, the elements can be formed integrally with each other, or may be formed separately and put together. Furthermore, “operatively coupled” can mean the element is directly coupled to the other element, or intervening elements may be present between the elements. Furthermore, “operatively coupled” may mean that the elements are detachable from each other, or that they are permanently operatively coupled together.

[0088] Although specific embodiments have been illustrated and described herein, those of ordinary skill in the art appreciate that any arrangement which is calculated to achieve the same purpose may be substituted for the specific embodiments shown and that the invention has other applications in other environments. This application is intended to cover any adaptations or variations of the present invention. The following claims are in no way intended to limit the scope of the invention to the specific embodiments described herein.

Claims

1. An anti-ligature hinge comprising: a first leaf; a second leaf operatively coupled to the first leaf; and an anti-ligature cap comprising: a body; one or more cap projections operatively coupled to the body or one or more cap recesses formed by the body; and a cap leaf operatively coupled to the body; wherein the one or more cap projections or the one or more cap recesses are operatively coupled to the first leaf or the second leaf.

2. The anti-ligature hinge of claim 1, wherein the anti-ligature cap comprises the one or more cap projections.

3. The anti-ligature hinge of claim 1, wherein the first leaf or the second leaf has a length shorter than the other to allow for the first leaf and second left to close with respect to the cap leaf.

4. The anti-ligature hinge of claim 1, wherein the first leaf and the second leaf comprise: a web;

and a plurality of knuckles having apertures therethrough; wherein a pin operatively couples the first leaf and the second leaf through the plurality of knuckles of the first leaf and the second leaf.
5. The anti-ligature hinge of claim 4, wherein the anti-ligature cap comprises the one or more cap projections, and wherein the one or more cap projections comprise a knuckle projection, wherein the knuckle projection extends into at least one knuckle of the plurality of knuckles of the first leaf or the second leaf.
6. The anti-ligature hinge of claim 1, wherein the anti-ligature cap comprises the one or more cap recesses formed by the body, wherein the one or more cap recesses are configured to receive one or more hinge projections operatively coupled to the first leaf or the second leaf.
7. The anti-ligature hinge of claim 1, wherein the first leaf or the second leaf have an upper end with a notched section.
8. The anti-ligature hinge of claim 7, wherein the notched section comprises a beveled surface that extends into at least a portion of the upper end of the first leaf or the second leaf.
9. The anti-ligature hinge of claim 8, wherein the beveled surface has an angle from a horizontal plane of an upper door edge that ranges from 10 to 80 degrees, inclusive.
10. The anti-ligature hinge of claim 7, wherein the body of the anti-ligature cap comprises: a bottom surface; wherein the one or more cap projections extend from the bottom surface of the body or the one or more cap recesses are formed within the bottom surface of the body.
11. The anti-ligature hinge of claim 1, wherein the body of the anti-ligature cap comprises: a top surface comprising of an exposed portion and a frame portion, wherein the frame portion is located within a door frame when the anti-ligature hinge is installed.
12. The anti-ligature hinge of claim 11, wherein the exposed portion diverges from the frame portion outwardly and downwardly from a horizontal plane of the frame portion.
13. The anti-ligature hinge of claim 12, wherein the exposed portion diverges from the frame portion at an anti-ligature angle from the horizontal plane of the frame portion that ranges from 10 to 80 degrees, inclusive.
14. The anti-ligature hinge of claim 1, wherein the body of the anti-ligature cap comprises: a first side cap surface; and a second side cap surface; wherein the cap leaf is flush with the first side cap surface and the cap leaf is flush with a first outer surface of the first leaf and the second side cap surface is flush with a second outer surface of the second leaf when the first leaf and the second leaf are in a closed position.
15. The anti-ligature hinge of claim 1, wherein the anti-ligature cap is a cast anti-ligature cap.
16. The anti-ligature hinge of claim 1, wherein the anti-ligature hinge is a continuous anti-ligature hinge.
17. A door system comprising: a door; a door frame; and an anti-ligature hinge comprising: a first leaf; a second leaf operatively coupled to the first leaf; and an anti-ligature cap comprising: a body; one or more cap projections operatively coupled to the body or one or more cap recesses formed by the body; and a cap leaf operatively coupled to the body; wherein the one or more cap projections or the one or more cap recesses are operatively coupled to the first leaf or the second leaf.
18. The door system of claim 17, wherein the body of the anti-ligature cap comprises: a first side cap surface; and a second side cap surface; wherein the cap leaf is flush with the first side cap surface and the cap leaf is flush with a first outer surface of the first leaf and the second side cap surface is flush with a second outer surface of the second leaf when the first leaf and the second leaf are in a closed position.
19. The door system of claim 17, wherein the anti-ligature cap comprises the one or more cap projections, and wherein the one or more cap projections comprise a knuckle projection, wherein the knuckle projection extends into at least one knuckle of a plurality of knuckles of the first leaf or the second leaf.
20. A method of installing an anti-ligature hinge comprising: assembling one of a first leaf or one of a second leaf to a door frame; assembling another of the first leaf or the second leaf to a door;

and assembling an anti-ligature cap to the first leaf or the second leaf, wherein the anti-ligature cap comprises: a body; and one or more cap projections operatively coupled to the body or one or more cap recesses formed by the body; a cap leaf operatively coupled to the body: wherein the one or more cap projections or the one or more cap recesses are operatively coupled to the first leaf or the second leaf.